MLDM Assignment 6

Detailed results for Jandson Ribeiro

Course	**Machine Learning and Data Mining WS 2022/23 (copy)**
Test	**MLDM Assignment 6**

This are your test results

Knowledge Questions

kNN

Status	Seen but not answered	
Your score	0 / 6	0%

Response

Select Right/Wrong for each statement.

Note:

- **Correct answer: 1 point**
- **Incorrect answer: -0.5 point**
- **Unanswered: 0 points**
- **Max score: 6 points**
- **Min score: 0 points**

Unanswered	Right	Wrong	
⦿	○	○	The scale of the data does not affect KNN performance.
⦿	○	○	KNN can be used as a technique to handle the missing values of only continuous variables.
⦿	○	○	The hamming distance can be used as a distance metric in KNN for text classification.
⦿	○	○	KNN algorithm is an example of deductive learning.
⦿	○	○	In KNN, the label prediction on unseen data requires a model that is generalized on training data.
⦿	○	○	The prediction time on the unseen sample is slower than the learning time in KNN

Evaluation

Status	Not started		
Your score	0 / 6		0%

Response

Drag and drop the statements on the left side to their corresponding statements on the right side.

Note:

- **Correct answer: 1 point**
- **Incorrect answer: -0.5 point**
- **Unanswered: 0 points**
- **Max score: 6 points**
- **Min score: 0 points**

Type 1 error

Specificity

Accuracy

Type 2 error

Sensitivity

Precision

Out of all the predicted samples of a class,how many samples are predicted correctly.

A person is covid positive but the test claims that the person is not positive.

out of all the samples of a class how many samples are predicted correctly.

used in ROC curves.

Not a good evaluation metric for skewed class distribution.

A person is not covid positive but the test claims that the person is positive.

Bayesian classification

Status	Not started	
Your score	0 / 6	0%

Response

Select Right/Wrong for each statement.

Note:

- **Correct answer: 1 point**
- **Incorrect answer: -0.5 point**
- **Unanswered: 0 points**
- **Max score: 6 points**
- **Min score: 0 points**

Unanswered	Right	Wrong	
⊙	○	○	Naive Baye's classifier does not work well on correlated features.
⊙	○	○	The accuracy of Naive Baye's classifier is affected by the noise in test data.
⊙	○	○	In naive Bayes, smoothing techniques can handle missing values in the data.
⊙	○	○	The prior probabilities are the hyperparameters of Bayesian Classifier.
⊙	○	○	The change of prior probability does not affect the posterior probability in Bayesian statistics.
⊙	○	○	The 'curse of dimensionality' has no effect on the Bayesian classifier.

Essay

Status	Not started	
Your score	0 / 14	0%

Response

Explain overfitting and underfitting in the KNN algorithm with respect to the choice of **K.**Explain a technique to choose the optimal value of **K.**

0 word

Practice

Confusion matrix

Status	Not started	
Your score	0 / 8	0%

Response

The below table is a confusion matrix of a classifier with the predicted labels on the top and the actual labels on the left-hand side. Your task is to evaluate the performance of the classifier by computing the metrics that are asked below.provide your answer with two digits after the decimal point.

Note:

- **Correct answer: 1 point**
- **Incorrect answer: 0 points**
- **Unanswered: 0 points**
- **Max score: 8 points**
- **Min score: 0 points**

	Apple	Banana	Pear
Apple	33	8	9
Banana	6	40	4
Pear	18	5	27

Micro-average Precision=

Micro-average Recall=

Micro-average F1-score=

Macro-average Precision=

Macro-average Recall=

Macro-average F1-score=

Average Accuracy=

Error=

Likelihood

Status	Not started	
Your score	0 / 8	0%

Response

Table A contains the data related to people weight, age, and gender. Your task is to calculate the likelihood for the data given in table B and provide your answer in the box below with two digits after the decimal point.

Note: Use the Population Variance formula and also take two digits after the decimal point for mean and variance.

- **Correct answer: 4 points**
- **Incorrect answer: 0 points**
- **Unanswered: 0 points**
- **Max score: 8 points**
- **Min score: 0 points**

Table A

Weight	Age	Gender
57	25	male
98	20	female
65	19	male
76	56	male
54	42	female
68	27	female

Table B

Weight	Age	Gender
57	23	

P(weight|male) =

P(age|female)) =

▪▪▪ Bernoulli Naive Bayes

Status	Not started		
Your score	0 / 6		0%

Response

Table A contains the data on which Bernoulli Naive Baye's classifier is trained. Use the classifier to predict whether football can be played for the data in Table B. Calculate the probabilities and write your answers in the box given below. Provide your answer with two digits after the decimal point.

Table A

Rainy	Sunny	Windy	Play Football
Yes	No	No	No
Yes	No	Yes	Yes
No	Yes	No	Yes
Yes	Yes	Yes	Yes
No	No	No	No
No	Yes	Yes	No

Table B

Rainy	Sunny	Windy	Play Football
No	No	Yes	?

P(Play Football =Yes|(Rainy,Sunny,Windy)) =

P(Play Football =No| (Rainy,Sunny,Windy)) =

Play Football =

▪▪▪ kNN

Status	Not started		
Your score	0 / 8		0%

Response

Predict the label for the data in the last row of the table using the KNN classifier with Euclidean distance as the distance metric. Write your answers for the predicted label for different values of K in the box given below.

Height	Weight	Age	Income
5.3	67	22	high
5.7	76	27	high
5.11	58	26	high
5.1	62	21	low
4.2	67	23	low
5.6	89	24	high
5.8	80	27	??

When K=1,Income=

When k=3,Income=

··· Naive bayes with smoothing

Status	Not started	
Your score	0 / 8	0%

Response

Naive Baye's classifier is trained on the data in Table A.Use the trained classifier to predict the label for the data in Table B.Use Laplace smoothing for Zero probabilities. Calculate the probabilities and provide your answer with two digits after the decimal point in the box given below.

Table A

Fever	cough	cold	stomach ache	Covid
High	Yes	No	Severe	Yes
Mild	No	Yes	Mild	No
Low	No	No	Severe	No
Low	Yes	Yes	Severe	Yes
High	Yes	Yes	Mild	Yes
Low	No	Yes	Mild	No

Table A

Fever	Cough	Cold	Stomach ache	Covid
High	No	No	Mild	??

P(Covid=yes | (Fever=high ,Cough=No,Cold=No,Stomach ache=Mild))=

P(Covid=No | (Fever=high ,Cough=No,Cold=No,Stomach ache=Mild)) =

Covid=

Programming

kNN program

Status	Not started	
Your score	0 / 30	0%

Response

Download the file assignment6.xlsx and write the code to implement the KNN algorithm.

Predict the label for the data in the table below, when you choose K=3 and K=5 and K=7

sepal_length	**sepal_width**	**petal_length**	**petal_width**	**species**
8.2	**2.6**	**3.2**	**1.8**	**??**

Please upload your code (notebook) below.

Make sure to print the output in the notebook while uploading.

Note:

You are not allowed to use the **scikit-learn** library.

File

No file uploaded